

Math 301

Midterm Exam - **Solution**

November 6, 2023

Barcode

Group: ☐ IET/MET

☐ MCTR

☐ EDPT

☐ EMS

Instructions

1. The **duration** of this exam is **120 minutes**.
2. Calculators are allowed.
3. There are **6 questions** in total.
4. There are **11 pages** in total in this booklet. Limit your **answers** to pages **2-8**. Use pages **9-11** as **scratch** papers.
5. **For each question**, read the given **instructions** carefully before you start providing your answers.
6. **For full credit**, let your **answers** be **detailed** to the best of your knowledge. Straight **calculator answers** will result in significant **reduction** of your score.
7. Do not spend too much time on any particular question. **Tackle the questions you feel comfortable answering first**.
8. **Indicate** clearly **question numbers** that you are solving.

Question	1	2	3	4	5	6	Total
Max. score	18	18	13	22	11	18	100
Score							

1. Parts (a) and (b) of this problem are independent. (18=12+6 pts)

(a) Find and sketch the domain of definition of the function $f(x, y) = \ln \left(\frac{y}{x^2 + y^2 - 1} \right)$.

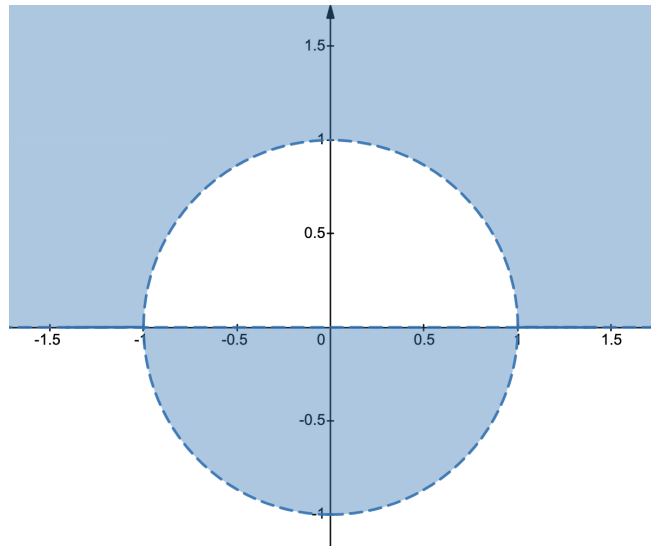
(b) Find $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y^2}{x^2 y^2 + (x+y)^2}$ if it exists or show that it does not exist.

Solution:

(a) $f(x, y)$ is defined if and only if:

$$(x^2 + y^2 - 1 > 0 \text{ and } y > 0) \quad \text{or} \quad (x^2 + y^2 - 1 < 0 \text{ and } y < 0).$$

Thus, the domain of f is represented by the shaded region in the below figure:



(b) • On the path $y = 0$ we have:

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{x \rightarrow 0} f(x, 0) = \lim_{x \rightarrow 0} 0 = 0.$$

• On the path $y = -x$ we have:

$$\lim_{(x,y) \rightarrow (0,0)} f(x, y) = \lim_{x \rightarrow 0} f(x, -x) = 1.$$

Since two different limits are obtained along two different paths, then the limit does not exist.

2. Parts (a) and (b) of this problem are independent. (18=9+9 points)

(a) Find $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$ and $\frac{\partial^2 f}{\partial y \partial x}$ of the function $f(x, y) = yx^2 \sin\left(\frac{x}{y}\right)$.

(b) i. Find the directional derivative of the function $g(x, y) = 1 + x\sqrt{2 + y^2}$ at the point $(0, \sqrt{2})$ in the direction $u = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$.

ii. Use linear approximation to estimate the value $g(0.01, \sqrt{2} + 0.02)$.

Solution:

$$\begin{aligned} \text{(a)} \quad \frac{\partial f}{\partial x}(x, y) &= 2xy \sin\left(\frac{x}{y}\right) + x^2 \cos\left(\frac{x}{y}\right), \\ \frac{\partial f}{\partial y}(x, y) &= x^2 \sin\left(\frac{x}{y}\right) - \frac{x^3}{y} \cos\left(\frac{x}{y}\right), \\ \frac{\partial^2 f}{\partial y \partial x}(x, y) &= \left(2x + \frac{x^3}{y^2}\right) \sin\left(\frac{x}{y}\right) - \frac{2x^2}{y} \cos\left(\frac{x}{y}\right). \end{aligned}$$

(b) i. The directional derivative:

$$D_u g(x, y) = \nabla g(x, y) \cdot u$$

$$\nabla g(x, y) = \left(\frac{\partial g}{\partial x}(x, y), \frac{\partial g}{\partial y}(x, y) \right) = \left(\sqrt{2 + y^2}, \frac{xy}{\sqrt{2 + y^2}} \right).$$

Therefore $\nabla g(0, \sqrt{2}) = (2, 0)$, and hence

$$D_u g(0, \sqrt{2}) = \nabla g(0, \sqrt{2}) \cdot u = (2, 0) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}} \right) = \sqrt{2}.$$

ii. The equation of the tangent plane to the graph of g at $(0, \sqrt{2}, 1)$ is given by:

$$\begin{aligned} T(x, y) &= g(0, \sqrt{2}) + \frac{\partial g}{\partial x}(0, \sqrt{2})(x - 0) + \frac{\partial g}{\partial y}(0, \sqrt{2})(y - \sqrt{2}) \\ &= 1 + 2x. \end{aligned}$$

Then

$$g(0.01, \sqrt{2} + 0.02) \approx T(0.01, \sqrt{2} + 0.02) = 1 + 0.02 = 1.02.$$

3. Parts (a) and (b) of this problem are independent.

(13=10+3 points)

(a) Assume that $z = f(x, y)$ has first partial derivatives and let

$$x = 2 + st, \quad y = s^2 + t^2.$$

Show the following matrix equation:

$$\begin{bmatrix} \frac{\partial z}{\partial s} \\ \frac{\partial z}{\partial t} \end{bmatrix} = \begin{bmatrix} t & 2s \\ s & 2t \end{bmatrix} \begin{bmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{bmatrix}.$$

(b) Let $z = x + g(u)$ where g is a differentiable function and $u = xy$. Show that

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = x.$$

Solution:

(a) By applying the chain rule we get

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial s},$$

with

$$\frac{\partial x}{\partial s} = t \quad \text{and} \quad \frac{\partial y}{\partial s} = 2s.$$

Therefore

$$\frac{\partial z}{\partial s} = t \frac{\partial z}{\partial x} + 2s \frac{\partial z}{\partial y}. \quad (1)$$

Similarly,

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t},$$

with

$$\frac{\partial x}{\partial t} = s \quad \text{and} \quad \frac{\partial y}{\partial t} = 2t.$$

Therefore

$$\frac{\partial z}{\partial t} = s \frac{\partial z}{\partial x} + 2t \frac{\partial z}{\partial y}. \quad (2)$$

Rewriting equations (1) and (2) in matrix form, we get

$$\begin{bmatrix} \frac{\partial z}{\partial s} \\ \frac{\partial z}{\partial t} \end{bmatrix} = \begin{bmatrix} t & 2s \\ s & 2t \end{bmatrix} \begin{bmatrix} \frac{\partial z}{\partial x} \\ \frac{\partial z}{\partial y} \end{bmatrix}.$$

(b) By applying chain rule we get

$$\frac{\partial z}{\partial x} = 1 + g'(u) \frac{\partial u}{\partial x} = 1 + yg'(xy),$$

and

$$\frac{\partial z}{\partial y} = g'(u) \frac{\partial u}{\partial y} = xg'(xy).$$

We now compute

$$x \frac{\partial z}{\partial x} - y \frac{\partial z}{\partial y} = x(1 + yg'(xy)) - y(xg'(xy)) = x + xyg'(xy) - xyg'(xy) = x.$$

4. **Parts (a) and (b) of this problem are independent.** (22=10+12 points)

- (a) The temperature at a point (x, y) on a metal plate is given by

$$T(x, y) = x^2 + y^2 + 1.$$

An ant on the plate walks around the ellipse of equation

$$\frac{x^2}{4} + y^2 = 1.$$

Use Lagrange Multipliers method to find the **highest** and **lowest** temperatures encountered by the ant.

- (b) Consider the function $f(x, y) = x^4 + y^4 - 2(x - y)^2$.

- i. Show that the critical points of f are:

$$(0, 0), \quad (\sqrt{2}, -\sqrt{2}) \quad \text{and} \quad (-\sqrt{2}, \sqrt{2}).$$

- ii. Show that the discriminant (Hessian) D of the function f is

$$D(x, y) = (12x^2 - 4)(12y^2 - 4) - 16.$$

- iii. Show that $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$ are local minima for f .

- iv. Find $D(0, 0)$. Answer by **yes** or **no** if the Second Derivative Test is appropriate to determine whether $(0, 0)$ is a local max/min or a saddle point?

- v. Show that $f(x, x) > f(0, 0)$ for $x \neq 0$.

- vi. Show that $f(x, 0) < f(0, 0)$ for $0 < |x| < \sqrt{2}$.

- vii. Deduce the nature of the critical point $(0, 0)$.

Solution:

- (a) The problem is to find the max/min values of the temperature function T subjected to the constraint

$$g(x, y) = \frac{x^2}{4} + y^2 - 1 = 0.$$

Using Lagrange Multipliers Method, we need to solve the following system

$$\begin{cases} \nabla T = \lambda \nabla g \\ g(x, y) = 0 \end{cases} \iff \begin{cases} 2x = \frac{\lambda x}{2} \\ 2y = 2\lambda y \\ \frac{x^2}{4} + y^2 - 1 = 0 \end{cases} \iff \begin{cases} x(4 - \lambda) = 0 \\ y(1 - \lambda) = 0 \\ \frac{x^2}{4} + y^2 - 1 = 0. \end{cases}$$

The solution to this system is given by

$$[(0, \pm 1); \lambda = 1] \quad \text{and} \quad [(\pm 2, 0); \lambda = 4].$$

We then compute

$$T(0, \pm 1) = 2 < 5 = T(\pm 2, 0),$$

and, as a conclusion:

- the highest temperature is 5 u.o.t
- the lowest temperature is 2 u.o.t.

- (b) i. The critical points of f are the solutions of the system

$$\nabla f = (0, 0) \iff \begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} \iff \begin{cases} 4x^3 - 4(x - y) = 0 \\ 4y^3 + 4(x - y) = 0. \end{cases}$$

Adding those two equations yields

$$x^3 = -y^3 \iff x = -y.$$

We substitute in the first equation to get

$$x^3 - 2x = 0 \iff x(x^2 - 2) = 0,$$

and hence $x = 0, \pm\sqrt{2}$. The critical points are then

$$(0, 0), \quad (\sqrt{2}, -\sqrt{2}) \quad \text{and} \quad (-\sqrt{2}, \sqrt{2}).$$

- ii. We compute the discriminant D at a point (x, y) :

$$D = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = (12x^2 - 4)(12y^2 - 4) - 16.$$

- iii. Since

$$D(\sqrt{2}, -\sqrt{2}) = D(-\sqrt{2}, \sqrt{2}) > 0 \quad \text{and} \quad \frac{\partial^2 f}{\partial x^2}(\sqrt{2}, -\sqrt{2}) = \frac{\partial^2 f}{\partial x^2}(-\sqrt{2}, \sqrt{2}) > 0,$$

we directly deduce that $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$ are points of local minimum for f .

- iv. Note that $D(0, 0) = 0$ hence the Second Derivative Test fails to determine the nature of $(0, 0)$.

- v. Direct computations yield

$$f(x, x) = 2x^4 > 0 = f(0, 0) \quad \text{for} \quad x \neq 0.$$

- vi. Direct computations yield

$$f(x, 0) = x^4 - 2x^2 = x^2(x^2 - 2) = x^2(x - \sqrt{2})(x + \sqrt{2}) < 0 = f(0, 0),$$

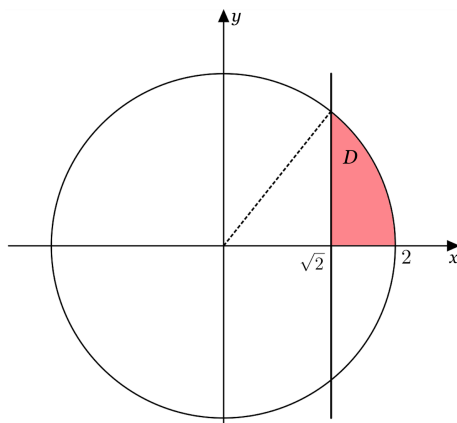
for $-\sqrt{2} < x < \sqrt{2}$ with $x \neq 0$.

- vii. Since for small $x \neq 0$, we have $f(x, 0) < f(0, 0) < f(x, x)$ then $(0, 0)$ is a saddle point.

5. The aim of this problem is to evaluate the double integral (11=4+4+3 points)

$$I = \iint_D y \sqrt{x^2 + y^2} dA,$$

where $D = \{(x, y) \in \mathbb{R}^2 : y \geq 0, x \geq \sqrt{2} \text{ and } x^2 + y^2 \leq 4\}$ is described in the figure below.



- (a) Give the equation of the line $x = \sqrt{2}$ and the circle $x^2 + y^2 = 4$ in polar coordinates.
 (b) By using polar coordinates, show that

$$I = \int_0^{\frac{\pi}{4}} \int_{\frac{\sqrt{2}}{\cos \theta}}^2 r^3 \sin \theta dr d\theta.$$

- (c) Deduce that value of I .

Solution:

- (a) Equations in polar coordinates of the straight line and the circle are respectively:

$$x = \sqrt{2} = r \cos \theta \rightarrow r = \frac{\sqrt{2}}{\cos \theta},$$

$$r = 2.$$

- (b) By using the polar coordinates, we have the following equality

$$\begin{aligned} I &= \iint_D y \sqrt{x^2 + y^2} dA = \int_0^{\frac{\pi}{4}} \int_{\frac{\sqrt{2}}{\cos \theta}}^2 r \sin \theta \sqrt{r^2 \cos^2 \theta + r^2 \sin^2 \theta} r dr d\theta \\ &= \int_0^{\frac{\pi}{4}} \int_{\frac{\sqrt{2}}{\cos \theta}}^2 r^3 \sin \theta dr d\theta. \end{aligned}$$

- (c) To evaluate I , we use part (b) and we start by computing

$$\int_{\frac{\sqrt{2}}{\cos \theta}}^2 r^3 \sin \theta dr = \left[\frac{r^4}{4} \sin \theta \right]_{\frac{\sqrt{2}}{\cos \theta}}^2 = 4 \sin \theta - \frac{\sin \theta}{\cos^4 \theta}.$$

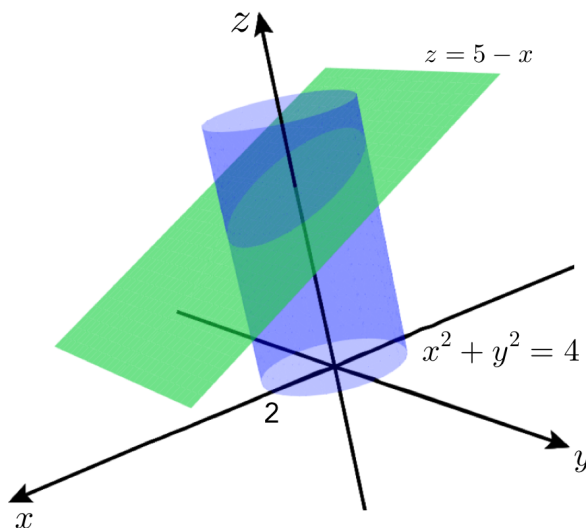
Therefore

$$\begin{aligned} I &= \int_0^{\frac{\pi}{4}} \left(4 \sin \theta - \frac{\sin \theta}{\cos^4 \theta} \right) d\theta = -4 \cos \theta \Big|_0^{\frac{\pi}{4}} + \int_0^{\frac{\pi}{4}} \frac{d(\cos \theta)}{\cos^4 \theta} \\ &= -2\sqrt{2} + 4 - \frac{1}{3 \cos^3 \theta} \Big|_0^{\frac{\pi}{4}} = \frac{1}{3}(13 - 8\sqrt{2}). \end{aligned}$$

6. Parts (a) and (b) of this problem are independent. (18=9+9 points)

(a) By using cylindrical coordinates, find the volume of the region (E) defined by:

$$x^2 + y^2 \leq 4 \quad \text{and} \quad 0 \leq z \leq 5 - x.$$



(b) Let (S) be the solid enclosed between the two spheres of equations:

$$x^2 + y^2 + z^2 = 1 \quad \text{and} \quad x^2 + y^2 + z^2 = e^2.$$

By using spherical coordinates, find the **mass** of (S), knowing that the density at each point (x, y, z) of (S) is given by $f(x, y, z) = (x^2 + y^2 + z^2)^{-\frac{3}{2}}$.

Solution:

(a) The volume of the region (E) is given by

$$\begin{aligned} \text{vol} &= \iiint_{(E)} dV = \int_0^{2\pi} \int_0^2 \int_0^{5-r\cos\theta} r \, dz \, dr \, d\theta \\ &= \int_0^{2\pi} \int_0^2 r(5 - r\cos\theta) \, dr \, d\theta = \int_0^{2\pi} \left(10 - \frac{8}{3} \cos\theta \right) d\theta = 20\pi. \end{aligned}$$

(b) The mass of the solid (S) is given by

$$\begin{aligned} \text{mass} &= \iiint_{(S)} f \, dV = \int_0^{2\pi} \int_0^\pi \int_1^e \rho^{-3} \rho^2 \sin\varphi \, d\rho \, d\varphi \, d\theta \\ &= \int_0^{2\pi} \int_0^\pi \sin\varphi \, d\varphi \, d\theta = 4\pi. \end{aligned}$$

End of Exam

Scratch Paper

Scratch Paper

Scratch Paper